RETROENGINE BETA ENCYCLOPEDIA

Volume 4: QA, Bug Testing & Field Guide — JDK 25 & LWJGL 3.4 Validation Protocols

1. EXECUTIVE QA OVERVIEW & BETA SYSTEM VERIFICATION

RetroEngine is entering its public Beta phase targeting **JDK 25** and **LWJGL 3.4**. This Field Guide provides bug testers, graphics engineers, and gameplay developers with rigorous verification protocols and baseline expected behaviors across all core subsystems. LWJGL 3.4 introduces modernized FFI native bindings, updated GLFW 3.4 windowing primitives, and enhanced Vulkan/Metal memory allocator hooks, requiring precise validation of native memory boundaries, Virtual Thread locks, and 60 Hz physics solver sub-stepping.

Subsystem Domain	Core Specification	Target Runtime	Primary QA Focus
Runtime Core	60 FPS Frame Loop / Spin-Wait	`JDK 25 + LWJGL 3.4`	Sub-ms spin-wait frame pacing & zero GC pauses
Windowing & Input	`GLFW 3.4 Window Manager`	`LWJGL 3.4 Native`	Multi-monitor DPI scaling & raw mouse input
Graphics RHI	Vulkan 1.3 & Metal 3 RHI	`LWJGL 3.4 / MoltenVK`	Dynamic rendering & bindless descriptor stability
Audio System	OpenAL Soft 1.24+ Spatial	`LWJGL 3.4 OpenAL`	Distance attenuation & zero-allocation UI sound pool
Script Engine	In-Memory Janino Hot-Reload	`JDK 25 Virtual Threads`	Classloader isolation & hot-loop exception capture

Beta QA Rule: Zero-Allocation Tick Loop Telemetry

All engine tick loops (PhysicsSystem, SceneRenderPipeline, GameUIRenderer, UIGeometryBatch) must maintain zero garbage-collection allocations during gameplay simulation. Heap profiling in JDK 25 / LWJGL 3.4 should show flat memory graphs during active play.

2. RUNTIME FRAME LOOP & LWJGL 3.4 NATIVE TELEMETRY

The engine loop (EngineLoop / RuntimeLoop) runs a 60 FPS sub-millisecond spin-wait target using GLFW.glfwPollEvents() and GLFW.glfwSwapBuffers() under LWJGL 3.4. Frame timing uses a rolling 8-frame delta smoothing window to prevent time-step jitter.

Telemetry Metric	Expected Baseline	Anomaly / Bug Condition	Diagnostic Remediation
Frame Pacing	16.66 ms ± 0.5 ms per frame	Frame spikes > 33 ms or micro-stutter	Inspect background streaming tasks & GC pause logs
Spin-Wait Target	Sub-ms GLFW poll wait	Thread starvation under JDK 25	Verify Virtual Thread pool allocation bounds
LWJGL 3.4 GLFW Poll	Native event queue sweep < 0.2 ms	Input lag or missed key release events	Check HotkeyRouter chord debouncing & focus locks
Delta Smoothing	8-frame moving average (0.0166s)	Delta explosion on window move/resize	Confirm GLFW window pause lock clamps delta to 0.1s

LWJGL 3.4 Native Stack Pointer Allocations

Native calls requiring temporary struct buffers (e.g. GLFWvidmode, VkExtent2D) must use LWJGL 3.4's MemoryStack.stackPush() try-with-resources idiom. Any unclosed stack allocations will cause native stack corruption.

3. SCRIPT ENGINE HOT-RELOAD & EXCEPTION ISOLATION QA

Gameplay logic authored via EntityScript is dynamically compiled in-memory using Janino's SimpleCompiler. When script source files under Assets/Scripts/ are modified, ScriptEngine invalidates the previous classloader, recompiles bytecode in-memory, reinstantiates script instances, and re-injects propertyOverrides from ScriptComponent.

Test Scenario	Trigger Action	Expected Beta Behavior	Failure / Bug State
Live Hot-Reload	Modify .java file on disk during play	Script re-compiles < 1s; state preserved	Script stalls or duplicate instances spawned
Compile Error	Save script with syntax error	Error logged to Console panel; old script kept	Engine crash or unhandled Janino compiler exception
Runtime Exception	Trigger NullPointerException in onUpdate	Script instance disabled; stack trace logged	Engine tick loop crashes or frame freezes indefinitely
Property Re-Injection	Edit @InspectorProperty field during play	Values updated live via propertyOverrides	Field resets to default class value on hot-reload

Hot-Loop Failure Cache

Failed script compilations are recorded in ScriptEngine.failedScriptCache. Testers should verify that correcting the syntax error on disk automatically clears the failure entry and resumes script execution cleanly.

4. ECS ENTITY HANDLE & STATE BACKUP MANAGER QA

Entities are referenced via 64-bit generational handles (EntityHandle combining a 32-bit index and 32-bit generation count). When entering Play mode, EntityStateBackupManager serializes all active entity transforms, rigidbodies, and script properties. Upon pressing Stop, the backup manager restores all entity states in-place.

Verification Domain	Target Mechanism	Expected Operational Result	Edge-Case Anomaly
Generational Indexing	Entity creation & destruction cycles	Index recycled; generation incremented	Stale handle accesses wrong newly spawned entity
State Backup	Press Play -> Move entities -> Press Stop	Transforms & components restored exactly	Transform drift or dangling script state after stop
Component Pooling	Dense ComponentPool allocation	Linear memory layout; zero pointer chase	Sparse array fragmentation causing GC pauses
Hierarchy Reparenting	Drag entity into child in Hierarchy panel	Parent-child world transforms re-calculated	Cyclic parent assignments (A child of B, B child of A)

Cycle-Safe Re-parenting Validation

Testers must verify that isCycleSafeParent() blocks illegal drag-and-drop operations in the Hierarchy panel, preventing infinite recursion during scene transform hierarchy updates.

5. ASSET STREAMING & VRAM REFERENCE SWEEPER QA

The AssetStreamer leverages JDK 25 Virtual Threads for non-blocking disk I/O, Assimp model importing, and ImageIO decoding. Streamed meshes and textures are uploaded to GPU memory via LWJGL 3.4 buffers. Assets are reference-tracked by frame stamps; unreferenced VRAM assets are automatically swept after 600 frames (~10 seconds at 60 FPS).

Asset Type	Streaming Pipeline	Eviction Threshold	QA Verification Test
Texture Assets	ImageIO / STB Image -> LWJGL 3.4 Buffer	Unreferenced > 600 frames	Unload scene; verify GPU memory decreases by texture size
3D Mesh Models	Assimp Native -> MeshBufferData	Unreferenced > 600 frames	Spawn 100 prefabs, delete them, confirm VRAM release
Pre-Cooked Rpack	Direct binary swizzled .rpack reader	Immediate on scene swap	Verify level load times under 500 ms for 50MB rpack
Audio Clips	STB Vorbis / WAV -> OpenAL Buffer	Manual asset unload hook	Trigger 50 simultaneous 3D sounds; check OpenAL channels

LWJGL 3.4 Buffer Native Alignment

All mesh index and vertex buffers uploaded via LWJGL 3.4 must be allocated using MemoryUtil.memAlloc() or direct ByteBuffers to guarantee alignment for Vulkan 1.3 / Metal 3 staging copy queues.

6. 60 HZ PHYSICS SOLVER & COLLIDER INTERSECTION QA

The PhysicsSystem steps at a fixed 60 Hz interval (0.0166s) with a maximum of 2 sub-steps per frame (MAX_PHYSICS_STEPS = 2). Solvers evaluate dynamic AABB BVH broadphase trees, collision manifolds, Vice City character ground snapping, 4-wheel vehicle suspension, and trigger volume callbacks.

Physics Subsystem	Solver Mechanics	Expected Behavior	Anomaly / Tunneling Bug
Broadphase BVH	Dynamic AABB tree + SAH balancing	Spatial queries return candidate pairs < 0.5ms	Missed collision pairs when objects move fast
Narrowphase Manifold	Box, Sphere, Capsule separation	Positional impulse resolves overlap cleanly	Object jittering or sinking into static geometry
Character Controller	Ground ray snapping & stair step	Character climbs steps <= stepOffset smoothly	Character popping or getting stuck on slope limits
Vehicle Suspension	4-wheel raycast + Hooke's Law	Spring forces match stiffness & damping	Vehicle flipping uncontrollably on terrain seams
Trigger Volumes	isTrigger == true overlaps	Fires onTriggerEnter/Stay/Exit callbacks	Trigger callback misses entity or fires duplicate events

Collider Debug Wireframe Color Reference

Unselected active colliders render as GREEN (R:0.20, G:0.90, B:0.50). Selected colliders render as CYAN (R:0.35, G:0.75, B:1.00). Trigger colliders (isTrigger == true) render as YELLOW (R:0.95, G:0.85, B:0.20). Character ground probes render as GREEN/RED-ORANGE.

7. DUAL RHI HARDWARE VALIDATION (VULKAN 1.3 & METAL 3)

RetroEngine provides dual native graphics backends via LWJGL 3.4: VulkanBackend (using Vulkan 1.3 dynamic rendering and 4,096 bindless descriptor indexing) and MetalBackend (using Metal 3 argument buffers and Objective-C JNI bindings).

RHI Backend	Target OS & Hardware	Key Native Technology	QA Validation Requirement
VulkanBackend	Windows, Linux, macOS (MoltenVK)	VK_KHR_dynamic_rendering & Bindless	Verify zero VkRenderPass allocation; check validation layers
MetalBackend	macOS (Apple Silicon & Intel)	MTLArgumentEncoder & Metal 3 UMA	Confirm zero-copy shared memory allocations on M1/M2/M3
Shadow Pass	Cross-Platform Shadow Map	2048x2048 Depth32Float + 3x3 PCF	Check shadow acne / peter-panning at shadow map cascades
Shader Hot Reload	Live SPIR-V / MSL Swap	glslc compilation & runtime pipeline swap	Modify shader on disk; confirm background pipeline update

LWJGL 3.4 Vulkan Validation Layer Filter

During Beta testing, VulkanBackend must run with VK_LAYER_KHRONOS_validation enabled. Any validation layer error or warning logged to stdout indicates a pipeline layout or barrier mismatch that must be remediated.

8. OFFSCREEN COLOR-ID PICKING & UI BATCHING QA

Entity selection in the 3D Viewport uses a 32-bit integer GPU offscreen picking pass (pick_overlay / entity_id shaders). UI rendering is driven by UIGeometryBatch, which dynamically batches 2D quads, rounded cards, pills, and bitmap text glyphs into LWJGL 3.4 buffers.

Render Pass	Implementation	Bitmask / Buffer Structure	QA Verification Test
Color-ID Picking	32-bit Offscreen Pass	`0x0001+ Entity, 0xFFFE0001 Gizmo, 0x8000 Bone`	Click complex mesh; verify Hierarchy selects correct Entity
UI Batching	UIGeometryBatch Buffer	FloatBuffer vertex array + Atlas Texture	Render 1,000 UI elements; confirm single draw call
Font Rendering	UIFont Bitmap Atlas	UV mapping for ASCII glyph set	Verify crisp text rendering without blurry sampling artefacts
PiP Viewfinder	640x360 Camera Inset	Framebuffer texture blit to lower-right corner	Select Camera entity; verify live secondary scene view

Sub-Millisecond Readback Efficiency

The Color-ID picking pass reads back a single 1x1 pixel region at mouse coordinates. Testers should verify that viewport clicking does not stall the main render thread or cause frame drops during drag operations.

9. STANDALONE BUILD EXPORTER & RPACK COOKER QA

The StandaloneExporter builds native application bundles for Windows (.exe), macOS (.app), and Linux. Offline asset baking is executed by RpackCooker, which swizzles textures into GPU formats (BC3 / ASTC), pre-indexes 3D meshes, and package resources into optimized .rpack binary archives.

Target Platform	Executable Bundle	Embedded Runtime	QA Packaging Checklist
Windows 64-bit	`Executable (.exe)`	JDK 25 JRE + LWJGL 3.4 DLLs	Launch standalone build on clean Windows machine without installed Java
macOS Universal	`App Bundle (.app)`	JDK 25 JRE + LWJGL 3.4 dylibs	Verify Gatekeeper signing & Apple Silicon arm64 execution
Linux 64-bit	`Native Binary`	JDK 25 JRE + LWJGL 3.4 .so	Test execution under X11 & Wayland compositors
Rpack Cooker	`Binary Archive (.rpack)`	Compressed VRAM Packages	Verify pre-cooked rpacks load 5x faster than raw disk Assets/

Cross-OS Build Export Requirement

BuildExporterOverlay flags that building standalone executables for a specific target OS requires the corresponding platform JDK home (containing jmods/) and LWJGL 3.4 native libraries for that target architecture.

10. BETA TESTING CHECKLIST & BUG REPORTING FIELD PROTOCOL

When encountering anomalies during Beta testing, follow the standardized bug reporting checklist below before submitting an issue ticket to the engine development team. Comprehensive issue reports accelerate regression fixes.

Field Protocol Step	Action Item	Target Log / Evidence	Submission Requirement
1. System Audit	Record Host OS, CPU, GPU & Driver Version	DisplayService hardware query	Must state OS, GPU model, and JDK 25 / LWJGL 3.4 build
2. Console Capture	Open Console (~ key) and save log output	EngineConsolePanel buffer log	Attach complete console log including stack traces
3. RHI Identification	Confirm active graphics backend	Vulkan 1.3 vs Metal 3 status bar	Specify whether issue occurs on Vulkan, Metal, or both
4. Reproduction Steps	Write exact step-by-step reproduction sequence	Minimal test scene (.rscene)	Provide bulleted list starting from Project Launcher
5. Asset Isolation	Isolate failing model, script, or texture	Specific file under Assets/	Attach reduced reproduction asset package

Encyclopedia Conclusion & Summary

This completes Volume 4 of the RetroEngine Beta Encyclopedia. Volumes 1 through 4 provide a complete reference covering engine architecture, scripting, UI dockspace, RHI graphics engineering, and QA verification protocols.